



FACULTY GUIDE

Faculty Implementation Guide

Plan the assignment, build or select the game, test it, deploy it, and tell students exactly what counts.

Recommended order: Instructional goal -> assigned mode -> build, including Appearance -> review Readiness -> Generate -> test -> host game HTML -> LMS -> learner view -> Mastery Report -> review when warranted -> return to practice.

1. Define the instructional job

- Name the course unit or concepts the game supports.
- Decide whether the activity is review, retrieval practice, a fluency check, a challenge, or a completion assignment.
- Choose one primary mode for the assignment. Other modes can remain available for voluntary practice.
- Decide what students must submit: nothing, a result screenshot, a mastery-report screenshot, a verification code, a CSV, or another explicit course artifact.

2. Build the game with the current Composer

Use the Faculty Build Composer to select a course area and content. Appearance themes and optional custom images are validated and embedded, so faculty do not edit image paths or source code.

01	Concepts and scope Choose an area and concepts. Open Learning outcomes: Standard gives normal coverage, Brief essentials, and Full all outcomes. Checkboxes create Custom coverage. Single-outcome presets coincide.
02	Game details Name the activity and file.
03	Modes Enable only the modes that fit the assignment. Availability checks reflect the selected content and required inventory.
04	Checkpoints Review automatic coverage and add optional focus only when instructionally useful.
05	Appearance Choose an official theme or optional slot-by-slot official or custom artwork.
06	Readiness Resolve blocking coverage problems and review the estimated package size.

Download and archive the ZIP and composition recipe.

Mode choice at a glance

Faculty enable the modes that fit the assignment; students see only those enabled in the generated build.

- **Standard Campaign:** A 30-room campaign with checkpoints, milestones, setbacks, and browser-local save/resume. Use for the complete game experience.
- **Timed Trial:** A ten-minute run with a visible countdown and no checkpoints; the timer governs completion. Use for fluency practice after students know the material.
- **Exam Drill:** Adaptive nine-question sections with review and revision before checkpoint commitment; correctness stays hidden while a section is active. Use for section-based review and deliberate answer commitment.
- **Legendary Mode:** A 30-room campaign using mastery-level, multi-step questions with close choices and no normal remediation sequence. Use as a later challenge after ordinary mastery is stable.
- **Score Attack:** A scored run rewarding correct answers, speed, and streaks while penalizing misses and rapid guessing. Use for repeat practice and motivation, not first exposure.
- **Quiz:** A fixed run of 1-15 questions chosen by the student while faculty-selected concepts remain fixed. Use when assigning a specific-length classroom or homework run.
- **Unlimited Practice:** Adaptive practice repeats in 30-question cycles until the student ends the session and generates a report. Use for open-ended practice where students decide when to stop.
- **Trial by Graph:** A 10-, 15-, or 20-question run using only audited items in which the graph is required; only supported lengths appear. Use for concentrated graphical-analysis practice.
- **Fading Fortune:** A 10-, 15-, or 20-question run. Random incorrect choices fade at difficulty-based intervals and value falls 100 -> 75 -> 50 -> 25. Use for time pressure without a full-run clock.
- **Risk & Reward:** A 10-, 15-, or 20-question run with a 1,000-point bankroll. Students risk 10%, 25%, 50%, or all in before seeing each question. Use for strategic confidence-and-stakes decisions.

3. Test before hosting

- Open every enabled mode and answer enough questions to see ordinary progression.
- Miss questions intentionally to confirm setbacks and repair/bridge behavior.
- Verify Timed Trial runs until the ten-minute timer expires rather than ending at the 30-room boundary.
- Verify Exam Drill supports section navigation and revision, keeps correctness hidden during the section, and advances after checkpoint commitment. Artifact powers remain disabled.
- Test Standard Campaign and Legendary save/resume in the same browser.
- Open graph questions, enlarge them, and test desktop and mobile layouts.
- Inspect the selected Appearance on desktop and mobile. Confirm the gameplay background remains recognizable without competing with question, answer, status, or Guide text.
- Confirm hallways and configured Guide, Boss, Artifact, and mode artwork appear as intended. Review the estimated final size before downloading the package.
- Complete each enabled mode and inspect Mastery Report. Confirm that clean runs can omit a review and that meaningful mapped weaknesses can open a web-hosted Concept Review and return to practice.

4. Host the generated game HTML

Composer output: The generated HTML embeds runtime logic, graph images, selected official or custom artwork, and build-specific Concept Review routing. Publish the game HTML by itself; no separate visual-asset folder is required. Review PDFs remain in the Mastery Quests web library. Keep the recipe, composition manifest, and README as faculty build records.

01	Rename the game index.html Optional, but recommended for a clean site root URL.
02	Upload the game HTML Publish index.html at the site root or chosen game path. No sibling Concept Review folder is required.
03	Enable HTTPS publishing For GitHub Pages: Settings → Pages → Deploy from a branch → main → /(root).
04	Test the public URL Use a private window so you are testing the published file, not a local cache or editor preview.

5. Put it in your LMS

The public HTTPS URL is the game. Add it to your LMS as an external link, or embed it if your LMS and institution permit external iframe content. There is no second game-wrapper file required.

```
<iframe title="Mastery Quest game" src="https://YOURUSERNAME.github.io/YOUR-REPO/" width="100%" height="900" style="border:0;" allowfullscreen></iframe>
```

- For a link, add the public URL using your LMS's normal link or module tool. For an embed, paste the iframe into the LMS HTML/source editor if one is available.
- Adjust the iframe height if your LMS layout creates internal scrolling.
- Test the activity in the LMS learner/student view, when available, after publishing.
- If the LMS or institutional security settings block the iframe, use the same public HTTPS URL as an external course link instead.

6. Explain data, evidence, and follow-up clearly

- Composer-generated games ask for a name or initials, not a school email.
- Public polished games and the free manual template keep progress and activity data in the browser. Remote anonymous telemetry is OFF by default. The Composer offers an explicit optional operational telemetry build; operational telemetry is not research by default. Application telemetry does not intentionally include names, email addresses, LMS account identifiers, university identifiers or free-text responses. Learners choose whether to share local downloads through the course workflow.
- If you require a CSV, screenshot, verification code, or mastery report, state exactly what to submit and where.
- Mastery Report summarizes performance evidence and the strongest and weakest signals; it does not claim perfect diagnosis from every run.
- When a meaningful weakness maps to an available review, students can open that one-page Concept Review from the Mastery Quests web library, then return to targeted practice, Repair, or Bridge. A clean run may produce no review recommendation.
- Do not promise that local progress follows the student across devices. It does not.

7. Accessibility and course fit

- Test keyboard navigation and graph enlargement.
- Check mobile behavior in the same LMS context students will use.
- Provide an alternative when institutional accessibility needs or device limitations make the game unsuitable for a learner.
- Do not make speed the primary grading criterion unless speed itself is an aligned learning outcome.

8. Final launch checklist

- ☐ The game title and assigned mode match the LMS instructions.
- ☐ The public URL works in a private browser window.
- ☐ Every enabled mode was tested.
- ☐ Intentional misses and remediation were tested.
- ☐ Graphs and mobile layout were tested.
- ☐ The selected official or custom artwork displays correctly, and gameplay text remains readable.
- ☐ Configured hallways and Guide, Boss, Artifact, and mode images render where expected.
- ☐ The estimated final package size was reviewed for the intended deployment.
- ☐ The generated game HTML was opened and tested before assignment.
- ☐ The LMS link or embed opens correctly in the learner/student view, when available.
- ☐ The evidence students must submit is stated clearly.
- ☐ The privacy statement matches the current local-storage workflow.
- ☐ A clean copy of the ZIP and composition recipe is archived.
- ☐ A mapped Mastery Report recommendation opens the correct Concept Review from the Mastery Quests web library.
- ☐ Student directions explain the review-to-practice loop without promising a review after every run.

Current guidance: <https://masteryquests.org/how-to/composer/> | Learning outcomes:

<https://masteryquests.org/how-to/learning-outcomes/> | Manual authoring and validation:

<https://masteryquests.org/downloads/resources/manual-faculty-guide.html>